

SENTINEL-CPS

Functional Requirements Document (FRD)

Securing Cyber Physical Systems in a Remote IoT Lab: A Zero-Trust Edge Gateway Architecture Utilizing eBPF and AI Threat Detection

Project Name	Sentinel-CPS
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Primary Focus	Functional requirements for Zero-Trust gateway mediation, TV-as-a-Track execution, encrypted ESP-NOW telemetry, and eBPF/AI threat mitigation

Prepared for Master's Thesis Development

Document Control

This document updates the functional requirements for the current Sentinel-CPS Version 10.0 implementation direction. It preserves the approved thesis title while aligning the requirements with the TV-as-a-Track physical execution model, Port 8080 unified C2 application, analog light-sensor vehicle navigation, high-IOPS USB storage, and planned eBPF/AI mitigation pipeline.

Version	Date	Prepared By	Summary of Update
10.0	May 29, 2026	Tarique Chowdhury	Updated the FRD to reflect Sentinel-CPS v10.0 requirements: bare-metal Gateway execution from USB 3.0 storage, systemd-managed Flask C2 service on TCP Port 8080, Hub-and-Spoke ESP32 telemetry, TV-as-a-Track execution, light-sensor/PID autonomous navigation, and host-level eBPF/AI anomaly response.

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1. Introduction

This Functional Requirements Document (FRD) defines the operational behaviors required for the Sentinel Edge Gateway and its integrated Cyber-Physical System (CPS). The requirements describe the Gateway daemon, Hub-and-Spoke ESP32 communication model, TV-as-a-Track execution environment, structured logging pipeline, and security observability functions needed to support Phase 2 eBPF monitoring and Phase 3 local AI anomaly inference.

The document is written as a thesis implementation specification. Each requirement describes what the system must do, why the capability matters, and what evidence should be collected during validation. The requirements are intentionally scoped to the Sentinel-CPS prototype and are not intended to redesign the university enterprise network or become a general-purpose robotics product specification.

2. Functional Scope and Operating Assumptions

- **Approved thesis title:** Securing Cyber Physical Systems in a Remote IoT Lab: A Zero-Trust Edge Gateway Architecture Utilizing eBPF and AI Threat Detection
- **Gateway role:** The Raspberry Pi Gateway is the central enforcement, command, logging, and observability point for the CPS environment.
- **Execution model:** The Gateway application runs natively as a systemd-managed service, not inside Docker or another container abstraction, to preserve host-level eBPF visibility.
- **Primary interface:** The unified C2 web application is exposed on TCP Port 8080 only to approved endpoints through Hardware-Level Access Control (HLAC).
- **Physical substrate:** The Smart TV serves as the dynamic light-emitting track surface and authorized display endpoint, but it is not an administrative workstation.
- **Edge navigation model:** ESP32 vehicles use downward-facing analog light sensors and onboard PID logic; camera-based vehicle navigation and high-bandwidth video streaming are deprecated from the current scope.
- **Threat model:** The monitoring pipeline focuses on abnormal serial activity, injected or replayed telemetry, excessive command frequency, and DoS-like behavior at the Gateway-to-Hub chokepoint.

3. Core Gateway Operations (FR-G)

The central Python daemon operates natively through systemd to preserve direct host observability. It must provide reliable serial translation, safe command execution, structured logging, hardware surface reduction, and a unified Port 8080 web interface.

ID	Requirement	Functional Rationale	Validation Evidence
FR-G1	Serial Translation: The Gateway daemon must establish and maintain a continuous, bi-directional serial connection with the tethered Hub ESP32 through /dev/ttyUSB0 at 115200 baud.	Maintains the Gateway-to-Hub chokepoint required for safe command dispatch and observable serial behavior.	Serial connection test; successful read/write exchange; timeout/error handling logs.
FR-G2	Command Execution: The daemon must translate approved C2 instructions, including Lane Builder track updates, vehicle control states, and emergency STOP commands, into serial messages sent to the Hub ESP32.	Ensures that user intent is mediated by the Gateway rather than sent directly to the edge network.	Command-to-serial trace; Hub acknowledgement; application log entry in actions.csv.

ID	Requirement	Functional Rationale	Validation Evidence
FR-G3	Telemetry Aggregation: The daemon must run a non-blocking background listener that continuously reads incoming ESP-NOW telemetry forwarded by the Hub ESP32.	Allows vehicle state, sensor readings, and operational status to update without blocking C2 web requests.	Background listener test; telemetry parsing output; live dashboard update under load.
FR-G4	Structured High-IOPS Logging: The daemon must record command interactions in actions.csv and vehicle telemetry in telemetry.csv or an equivalent structured telemetry store. File writes must use thread-safe locking such as fcntl.flock. Logging must occur on high-throughput USB 3.0 storage rather than MicroSD media.	Produces audit evidence and AI training data while reducing write-endurance risk during high-volume eBPF and telemetry collection.	Log integrity review; concurrent write test; USB storage path verification; sample baseline dataset.
FR-G5	Hardware Surface Reduction: The Gateway OS must enforce udev or equivalent kernel-level policy that disables unused USB interfaces while preserving the approved Hub ESP32 serial path.	Reduces the risk of physical peripheral attacks against the bare-metal Gateway.	udev ruleset review; unauthorized USB device test; /dev/ttyUSB0 availability test.
FR-G6	Unified Web Dashboard and Digital Twin: The Gateway daemon must serve a unified Flask-based C2 web application on TCP Port 8080. The interface must include an interactive Lane Builder and a live Digital Twin telemetry display.	Provides a single controlled surface for track generation, command workflow, telemetry visualization, and review.	Port 8080 access test; Lane Builder output; Digital Twin telemetry refresh test; firewall confirmation.
FR-G7	Service Persistence and Recovery: The Gateway application must be managed by systemd as sentinel.service and recover to a safe default state after reboot or daemon restart.	Ensures predictable lab operation and avoids unsafe edge behavior after service interruption.	systemctl status evidence; reboot test; safe STOP/default-state verification.

4. Physical CPS Operations (FR-C)

The distributed ESP32 components operate under a strict Hub-and-Spoke model. The Gateway remains tethered to a Hub ESP32, while autonomous edge vehicles perform local light-sensor navigation and communicate operational state through encrypted ESP-NOW.

ID	Requirement	Functional Rationale	Validation Evidence
FR-C1	Analog Sensor Nodes: Edge vehicles using ESP32-WROVER-DEV or equivalent ESP32 platforms must operate as high-speed navigation nodes with downward-facing 12-bit analog light sensors such as phototransistors or LDRs. Camera-based navigation is deprecated from this version.	Reduces visual spoofing surface, bandwidth requirements, and latency while preserving autonomous navigation capability.	ADC sampling test; light/dark threshold calibration; camera payload exclusion check.
FR-C2	Onboard PID Control: Each edge vehicle must use local PID or equivalent closed-loop control to remain centered over the high-luminance track emitted by the Smart TV.	Keeps time-sensitive steering decisions local to the vehicle and reduces dependency on remote command latency.	Track-following test; PID parameter log; observed steering stability on TV substrate.

ID	Requirement	Functional Rationale	Validation Evidence
FR-C3	Encrypted Telemetry Transmission: Edge vehicles must transmit operational telemetry, including ADC arrays, steering/PID state, vehicle identifier, and status flags, to the Hub using connectionless ESP-NOW.	Provides low-latency telemetry without granting edge vehicles direct access to the enterprise network.	ESP-NOW packet capture/serial output; telemetry schema review; latency sample.
FR-C4	Hub Bridging: The tethered ESP32-D0WDQ6 Hub must act as a transparent serial-to-wireless bridge. It must receive ESP-NOW telemetry, print it to /dev/ttyUSB0, and broadcast Gateway-approved commands back to authorized edge nodes.	Maintains Gateway isolation from raw wireless traffic while preserving a controlled command and telemetry path.	Hub firmware review; serial forwarding test; edge command acknowledgement.
FR-C5	Wireless Cryptography: ESP-NOW traffic must use CCMP encryption with a hardcoded 16-byte Local Master Key (LMK) and approved peer identity configuration.	Rejects spoofed or unauthorized payloads from untrusted physical MAC addresses.	Unauthorized peer rejection test; LMK configuration review; encrypted telemetry verification.
FR-C6	Safe Edge State Handling: Edge vehicles must support a STOP or neutral motor state that can be triggered by the Hub/Gateway during normal shutdown, anomaly response, or communication loss.	Protects the physical environment from unsafe actuator behavior.	STOP command test; communication-loss test; motor neutral-state evidence.

5. Lab Subsystem Integrations (FR-S)

Sentinel-CPS integrates the Gateway, Bastion Host, Smart TV, and edge vehicles into a controlled educational lab environment. These requirements define how external lab subsystems support dynamic physical execution and remote educational use.

ID	Requirement	Functional Rationale	Validation Evidence
FR-S1	TV-as-a-Track Physical Execution: The physical test environment must use a horizontally mounted Smart TV as the light-emitting road substrate. The TV must access the Port 8080 Lane Builder display surface to render dynamic white-on-black tracks for the physical vehicles.	Enables repeatable, software-defined track generation without tape, wood, or mechanical barriers.	Track rendering test; TV orientation/photo evidence; vehicle response to emitted track.
FR-S2	Display Endpoint Segregation: The Smart TV must be authorized only for required web/display access and must not receive SSH or administrative privileges.	Preserves Zero-Trust separation between display functionality and Gateway management.	nftables rule review; Port 8080 allow test; Port 22 deny test from Display Node.
FR-S3	Remote Educational Sandbox: The system must support an isolated development workflow in which remote students reach a Windows Bastion Host through VPN/RDP to interact with sandboxed development microcontrollers while observing the Gateway dashboard.	Concentrates remote user access at a controlled ingress point rather than exposing the Gateway or edge nodes directly.	Bastion Host workflow test; RDP access evidence; separation between sandbox hardware and Gateway control path.

ID	Requirement	Functional Rationale	Validation Evidence
FR-S4	Lane Builder and Digital Twin Synchronization: Track generation, telemetry overlays, and vehicle status must remain synchronized in the unified C2 interface.	Allows the same web surface to act as command interface, physical reality generator, and observational dashboard.	Lane update test; telemetry overlay refresh; timestamp comparison between serial telemetry and display state.

6. Observability and Security Pipeline (FR-O)

The observability pipeline creates the evidence environment required for eBPF tracing and AI-assisted anomaly detection. These functions are essential to the thesis contribution because they connect normal CPS operation with active security response.

ID	Requirement	Functional Rationale	Validation Evidence
FR-O1	Syscall Tracing: The bare-metal Gateway architecture must allow bcc-tools eBPF C programs to attach to sys_enter_write and sys_enter_read events associated with the /dev/ttyUSB0 serial interface.	Provides host-level visibility into the serial chokepoint that is harder to spoof than application logs alone.	eBPF script attach test; sample syscall trace; mapping between command logs and serial writes.
FR-O2	Threat Inference Trigger: A localized AI anomaly detection model must evaluate live or replayed eBPF telemetry, command logs, and expected telemetry frequency to score abnormal behavior such as replay, injection, or DoS-like bursts.	Supports automated detection of malicious or unsafe behavior in the command/telemetry path.	Baseline dataset; anomaly score output; simulated attack trace; false-positive review.
FR-O3	Automated Mitigation: Upon a critical anomaly score, the Gateway daemon must issue an emergency STOP override to the Hub ESP32 and assert a daemon-controlled serial lock or equivalent severance mechanism around the read/write buffer.	Provides an active response path that protects physical devices before unsafe edge behavior escalates.	STOP/severance test under attack load; serial lock evidence; recovery procedure log.
FR-O4	Baseline Dataset Generation: The system must generate sufficient nominal command, telemetry, and syscall data from normal TV-as-a-Track operation to train or calibrate the lightweight anomaly model.	Ensures the AI module is grounded in expected PID steering behavior and normal ESP-NOW payload rates.	Baseline runbook; telemetry.csv samples; model calibration report.

7. Functional Validation Criteria

The FRD will be considered satisfied if the prototype demonstrates the following outcomes under controlled lab conditions:

1. **Gateway command path:** The Gateway can send approved commands to the Hub over /dev/ttyUSB0 and record the action in structured logs.
2. **Telemetry ingestion:** The Gateway can continuously receive and parse Hub-forwarded telemetry without blocking the C2 application.
3. **TV-as-a-Track operation:** The Lane Builder can render a high-contrast track on the Smart TV and an ESP32 vehicle can respond to the emitted light path using analog sensing and PID control.

4. **Access separation:** Authorized endpoints can access the required services, while the Display Node remains restricted to web/display functionality and unapproved endpoints are denied.
5. **Observability readiness:** Application logs, serial traces, and telemetry outputs can be correlated to establish nominal behavior.
6. **Threat response:** When abnormal command or telemetry behavior is detected, the Gateway can trigger STOP and serial severance behavior to protect the physical CPS.

8. Requirement Traceability Summary

Trace Area	Relevant Requirements	Related Thesis Artifact / Validation Link
Gateway mediation	FR-G1, FR-G2, FR-G3, FR-G6, FR-G7	System Architecture Document Sections 5 and 9; Scope Deliverable: Unified C2 Web Application
High-integrity logging and storage	FR-G4, FR-O4	System Architecture Document Section 4 and Appendix A; Scope Deliverable: Zero-Trust Gateway Infrastructure
Physical attack surface reduction	FR-G5, FR-S2	Security Requirements: Hardware and perimeter controls; Scope Assumptions: Display endpoint constraint
TV-as-a-Track execution	FR-C1, FR-C2, FR-S1, FR-S4	System Architecture Document Section 6; Thesis Synopsis: TV-as-a-Track execution layer
Encrypted Hub-and-Spoke telemetry	FR-C3, FR-C4, FR-C5, FR-C6	System Architecture Document Sections 3 and 5; Scope Deliverable: Dynamic Physical Substrate
eBPF/AI detection and response	FR-O1, FR-O2, FR-O3, FR-O4	System Architecture Document Section 7; Scope Deliverable: Active Threat Mitigation
Remote educational sandbox	FR-S3	Thesis Synopsis: Remote Educational Sandbox; Scope Deliverable: Remote Educational Sandbox

Final Functional Summary

The Sentinel-CPS functional baseline defines a complete remote CPS education workflow: users enter through a controlled Bastion Host, interact with a Port 8080 Gateway C2 surface, generate a dynamic TV-based track, observe and command ESP32 edge vehicles through a Hub-and-Spoke telemetry bridge, and rely on host-level eBPF/AI monitoring for anomaly detection and emergency mitigation. These requirements position the project as a security-focused CPS architecture rather than only a robotics demonstration.